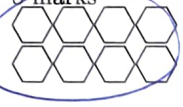


#	Pre-reqs	Milestone	
Setup the ROS 2 Package - Total 1 Marks			
1.1		Create a ROS 2 package with CMakeList.txt, package.xml, and all other required source files ready.	1 mark 
Complete the Class TB4ArcActionServer Definition - Total 4 Marks			
2.1	1.1	Complete the definitions of shared pointers for action server, command velocity publisher, and odometry subscriber.	1 mark 
2.2	2.1	Initialise the command velocity publisher shared pointer.	1 mark 
2.3	2.1	Initialise the odometry subscriber shared pointer, and bind the call back function "odom_callback"	1 mark 
2.4	2.1	Initialise the drive arc action server with name as "drive_arc_prac2", and bind call back functions for handling of accepting a goal, cancelling a action, and process the accepted goal	1 mark 
Complete the call back function odom_callback of receiving odometry message - Total 1 Marks			
3.1	2.3	Save the receive odometry message odom_msg to the class member variable odom_.	1 mark 
Complete all call back functions of receiving DriveArc action request - Total 3 Marks			
4.1	2.4	Complete the call back function handle_accepted that handles the accepted goal.	1 mark 
4.2	2.4	Complete the call back function handle_cancel that cancels the goal.	1 mark 
4.3	2.4	Complete the call back function handle_goal that accepts goal, you should also print the goal details in the terminal.	1 mark 
Complete the thread to process the accepted goal - Total 10 Marks			
5.1	4.1	Define variables to store action's goal, results, and feedback to be published.	1 mark 
5.2	4.1	Check whether action request is valid. If not valid, you should print a message to the terminal to tell the user an invalid request is received.	1 mark 
5.3	4.1, 2.2, 3.1	If the action request is valid. - calculate the command velocity signal based on received goal - decide the publication frequency, the times (or count) of publishing the command velocity - calculate the feedback information - publish the command velocity signal and feedback to the network - check whether Goal cancelling is received, and if yes, publish current feedback and result - when goal is achieved, publish the results	8 marks 

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